

system thinking & design

RAMAN SAXENA

Product Design & Management (PDM) and Software Engineering Research Centre (SERC) - IIIT Hyderabad

Lets Discuss what we learned from Web Activity:

- What did you notice when one person left the web?
- Did the effect stay local, or did it travel to people who weren't directly connected?
- Was there anything in the web that surprised you?
- Could you predict, before we pulled the strand, exactly how far the ripple would travel?

Class 1

Exercies 1 and Exercise 2

Examples of Systems

- Smartphone ecosystem
- Public transport
- Online learning platform

LECTURE 2

Lets Start from where we left (Exercies 2):

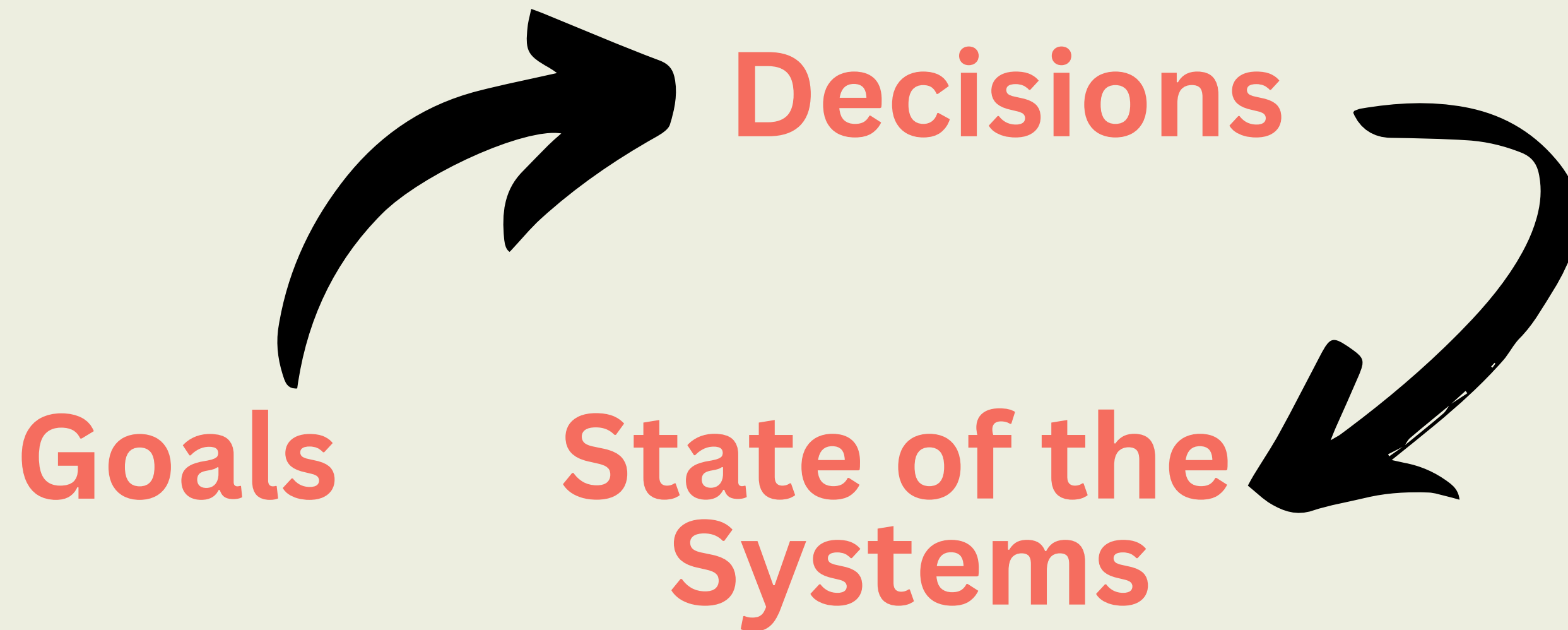
- "What happened when just one person moved?" ----> **Interconnectedness** (you can't change just one thing).
- "Could you control where the group ended up?" ----> **Emergent Behavior** (the system has a mind of its own).
- "Why was it hard to predict what would happen?" ----> **Non-Linear Dynamics** (small inputs can cause massive ripple effects).

Open Loop/Linear Thinking

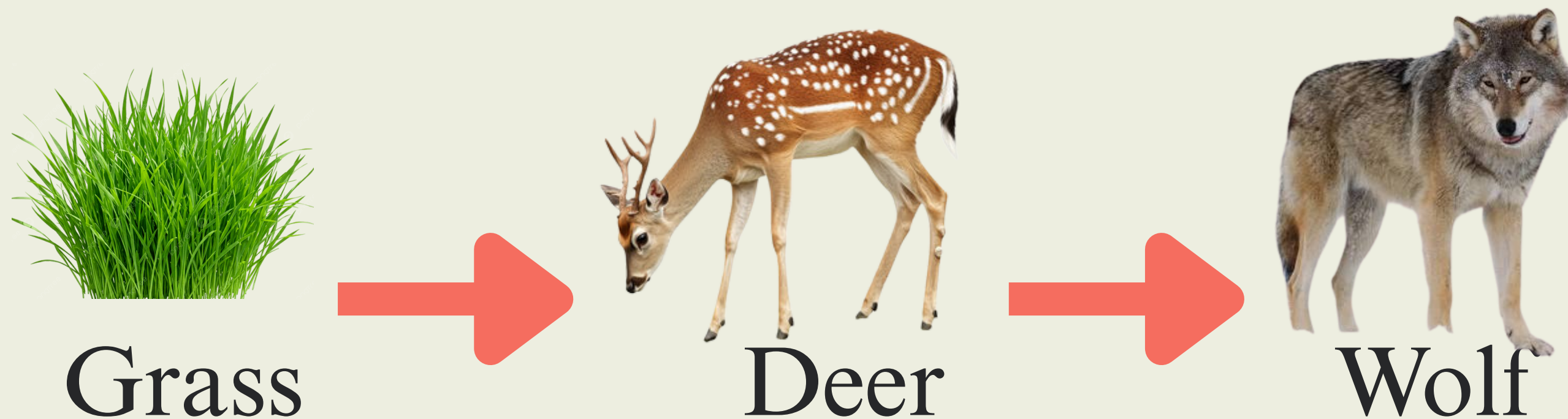


Open Loop/Linear Thinking is **not complete**

Open Loop/Linear Thinking

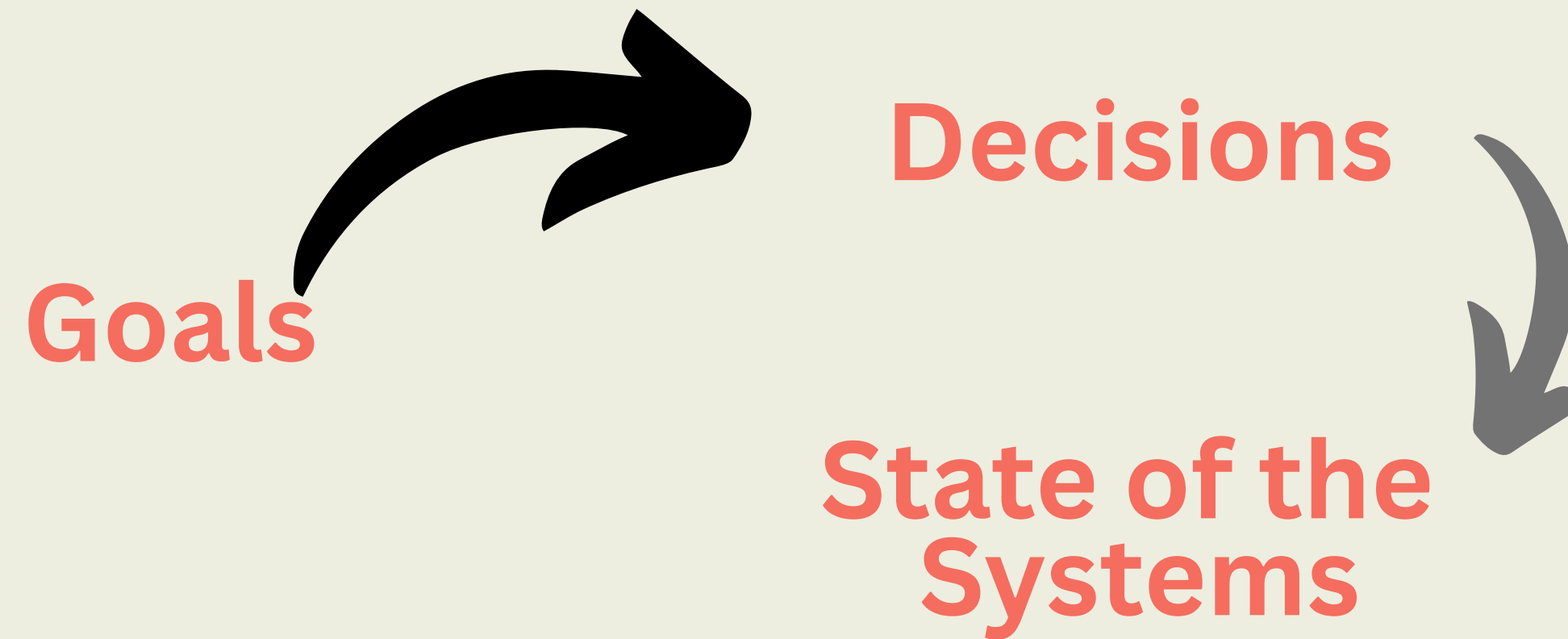


Linear thinking views a food chain as a simple string

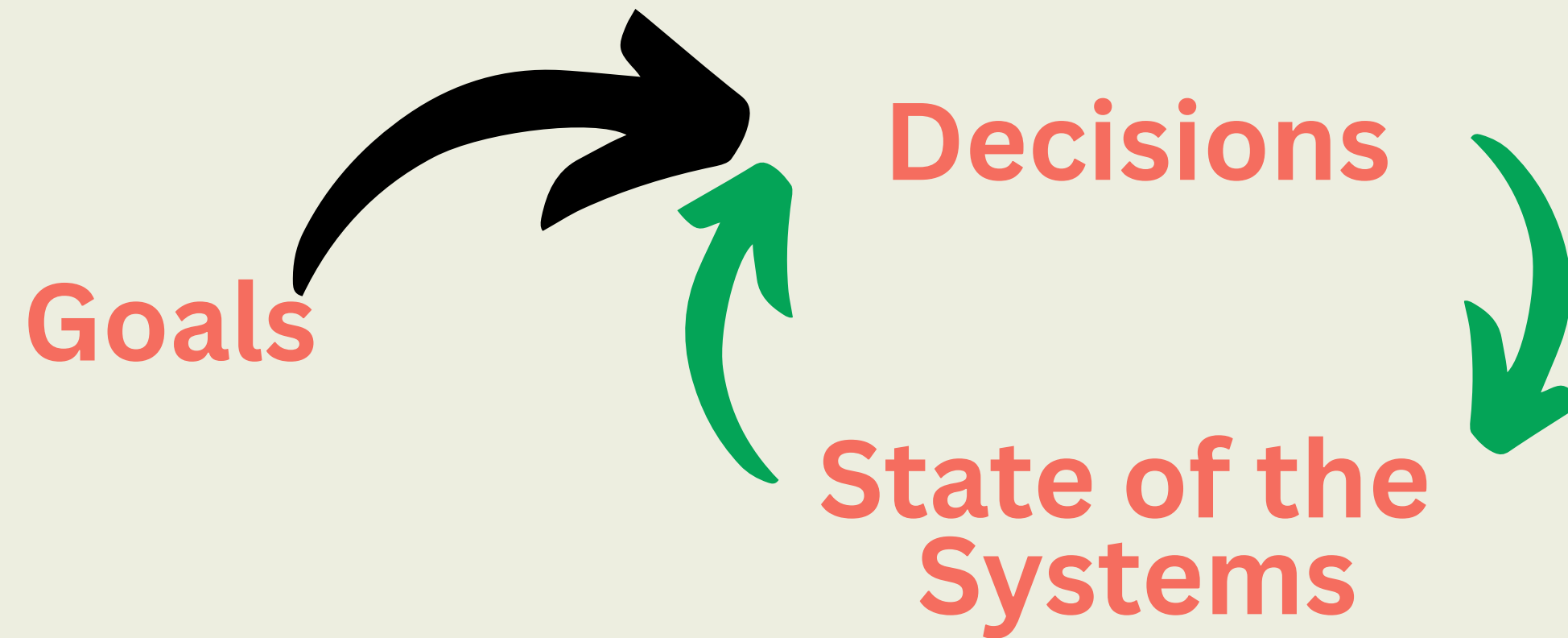


Open Loop/Linear Thinking is
not complete
because
We are embedded in a larger system

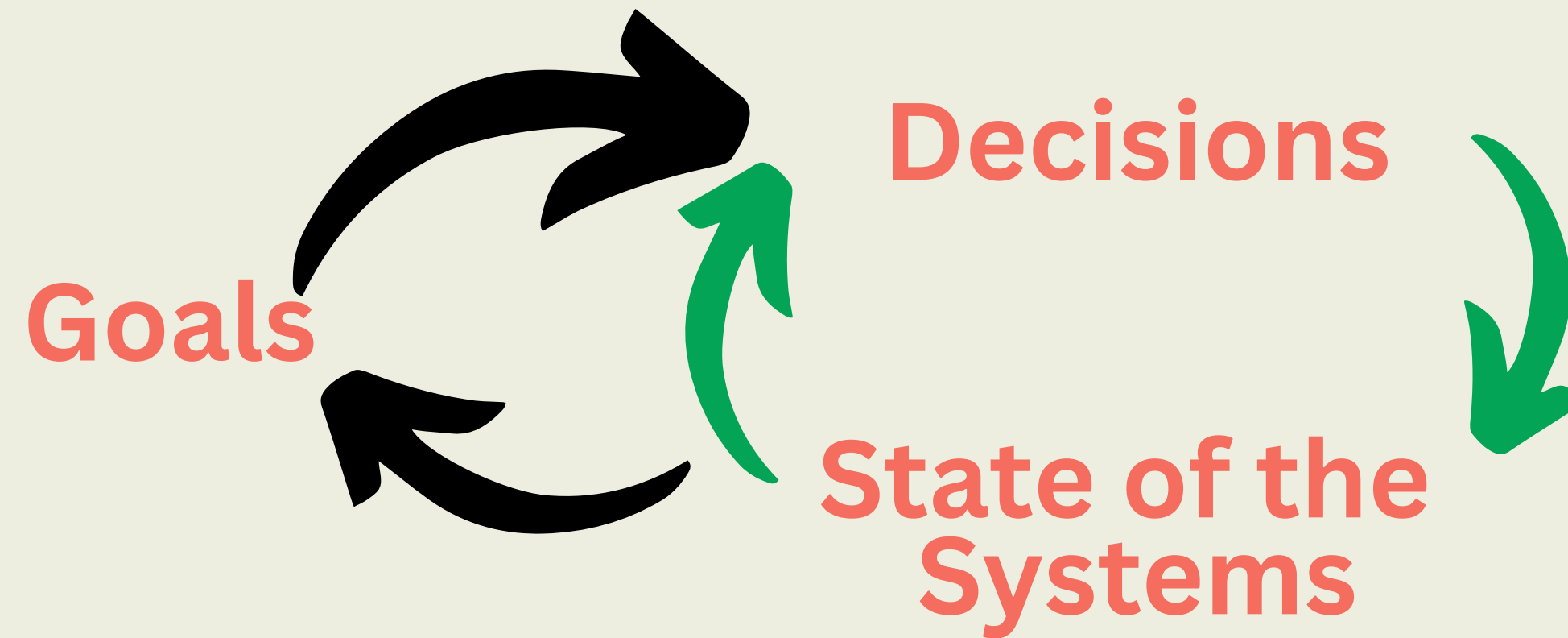
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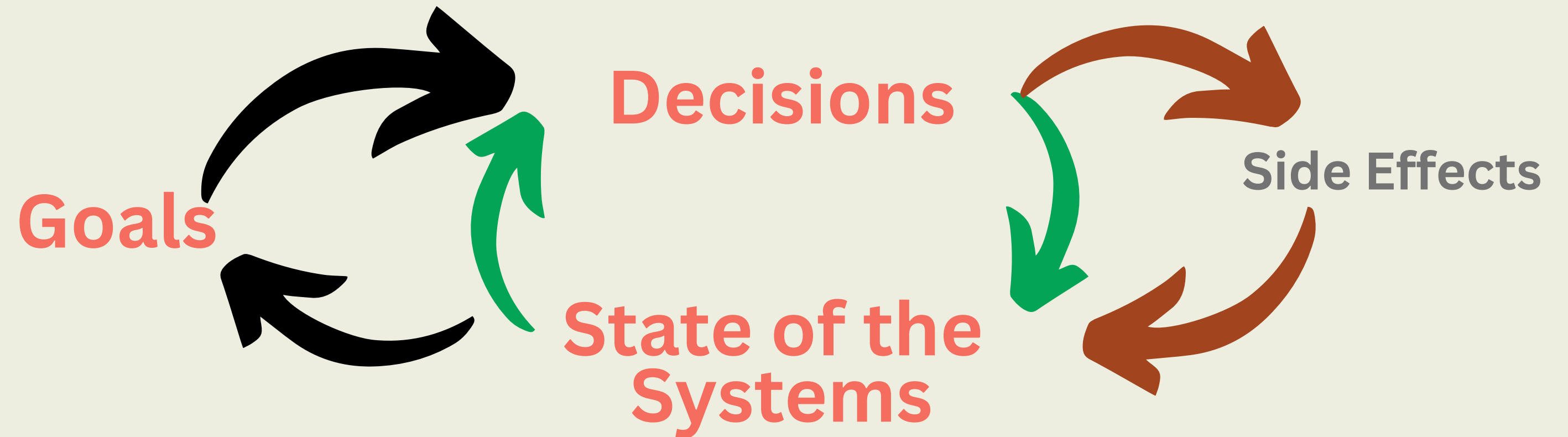
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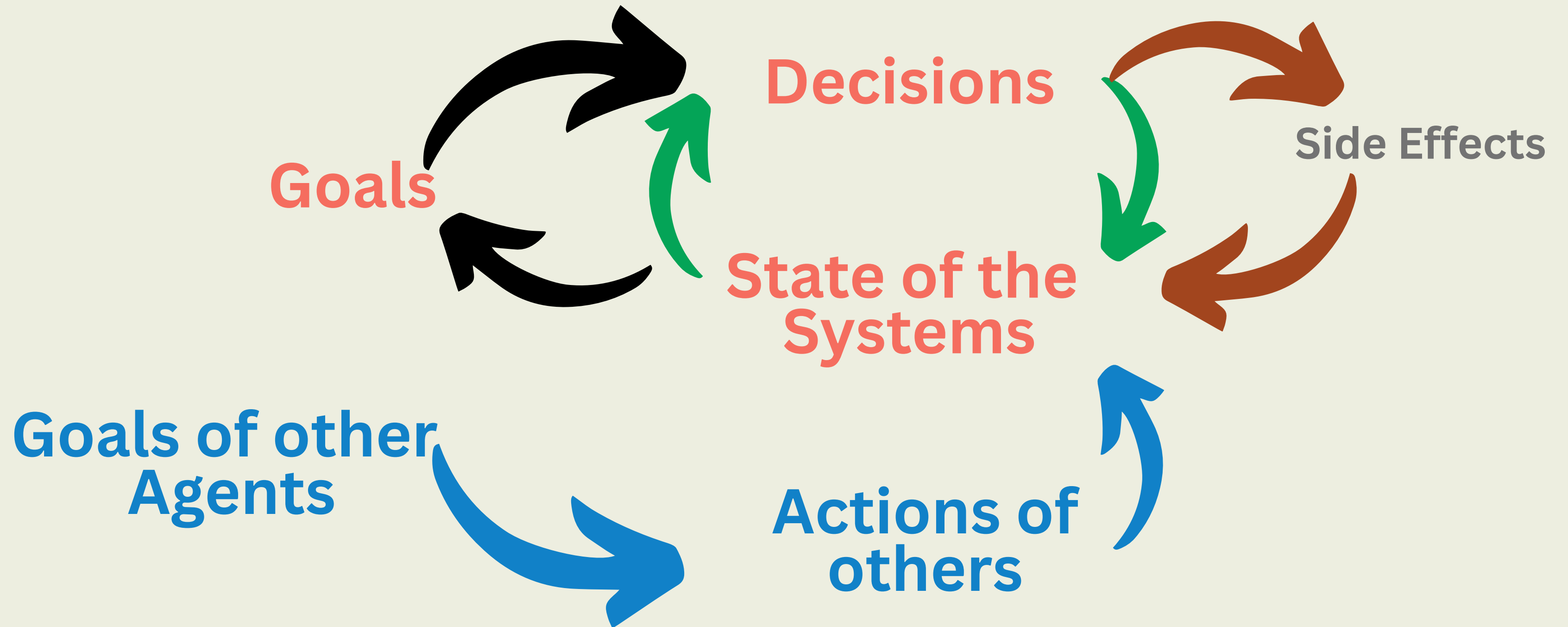
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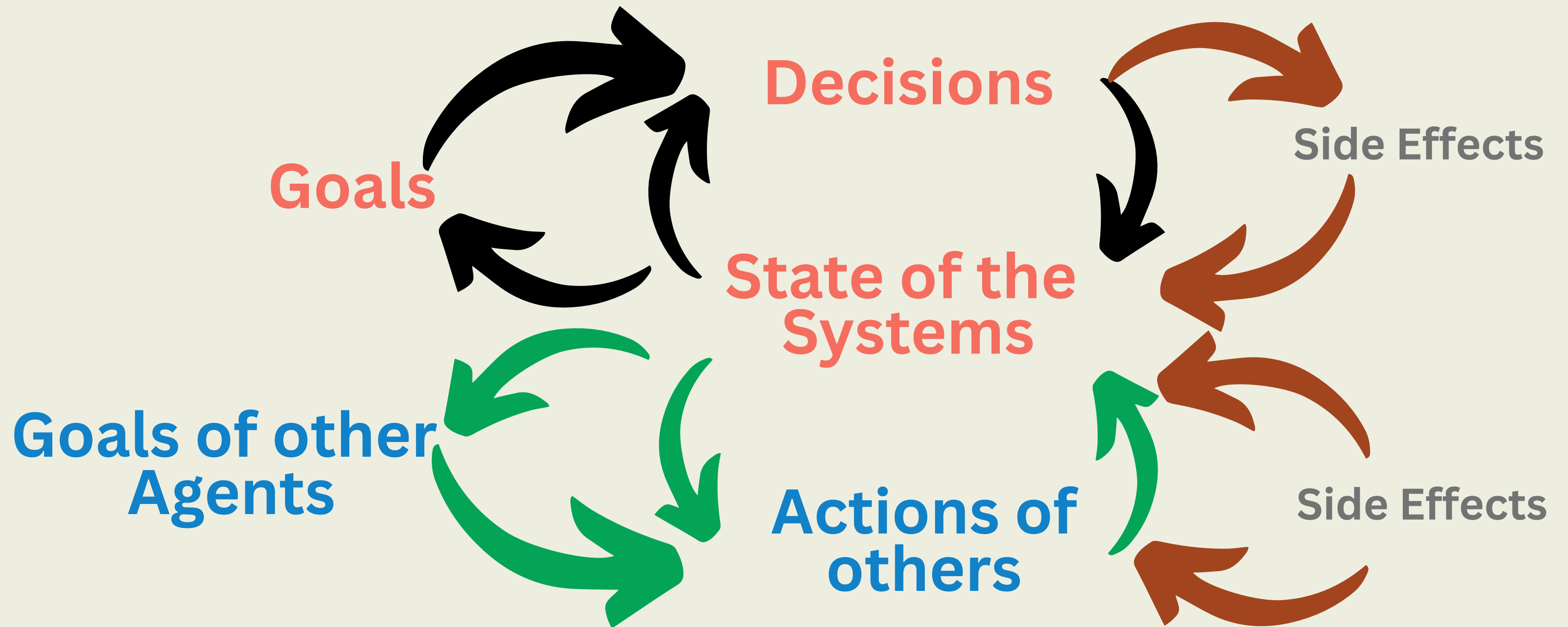
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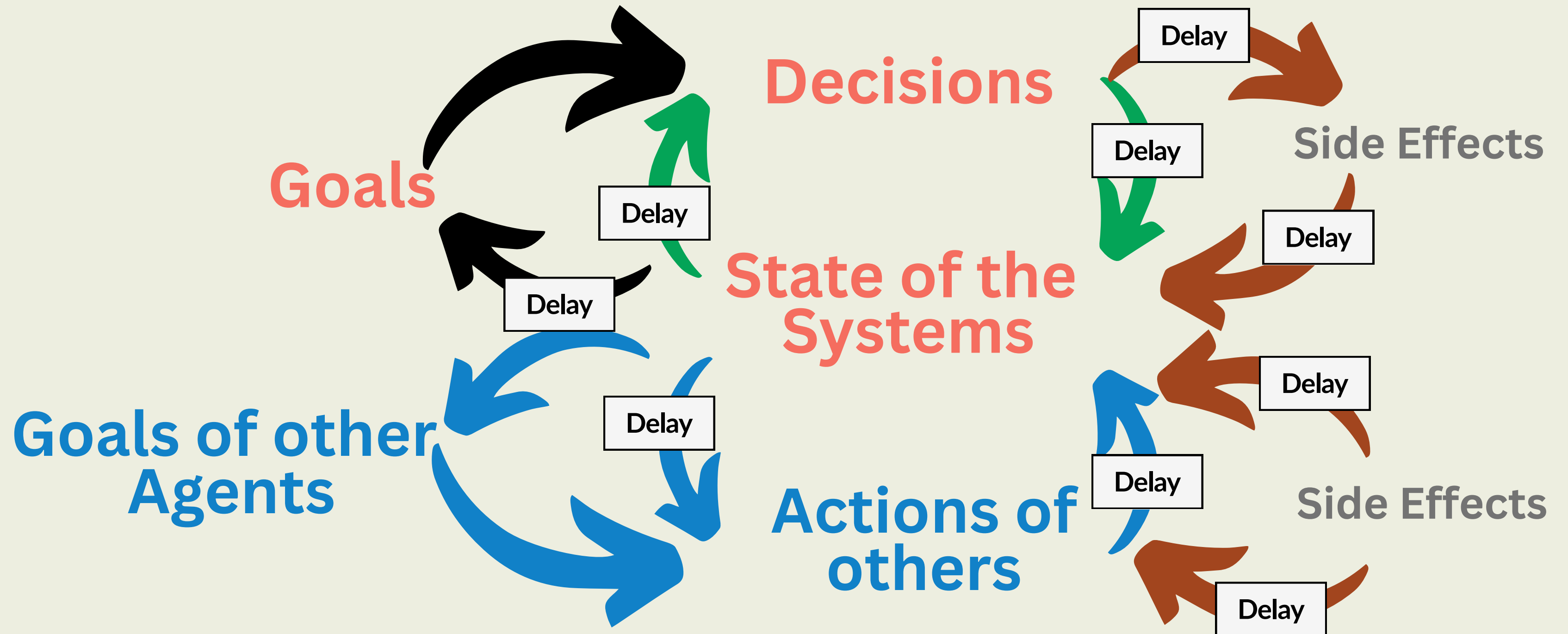
We are embedded in a **larger system**



We are embedded in a larger system



We are embedded in a larger system



System

A system is a set of things — people, cells, molecules, or whatever — interconnected in such a way that they produce their own pattern of behavior over time.

Donella Meadows, Thinking in Systems

Systems

“A system is an entity with interrelated and interdependent parts; it is defined by its boundaries and it is more than the sum of its parts (subsystem). Changing one part of the system affects other parts and the whole system, with predictable patterns of behavior.”

Source: Wikipedia – Systems Theory

What is a system?

A system is a set of things interacting in a way that produces something greater than the sum of its parts.

We live in a world of complex systems. A phone is a system. A rainforest is a system. This school is a system. You are a system. There are systems within systems: a community, within a city, within a country.

When we look at a system, we try to understand how all the parts work together. More importantly, we look at how a system interacts with, and influences, other systems.

Humans use systems as a way to understand the complexity of the world and to solve problems. Knowing our role in the systems around us allows us to function more effectively and proactively within them. And, understanding them allows us to anticipate outcomes that shape our lives.

Lets do an activity

We Need 2 Volunteers

- One Hold a Slinky on Hand
 - Other Hold the Paper Cylinder on Hand
- Apply a Push One by One

What Makes a Slinky Bounce Where as the Cylinder just fell over?

The Linear View

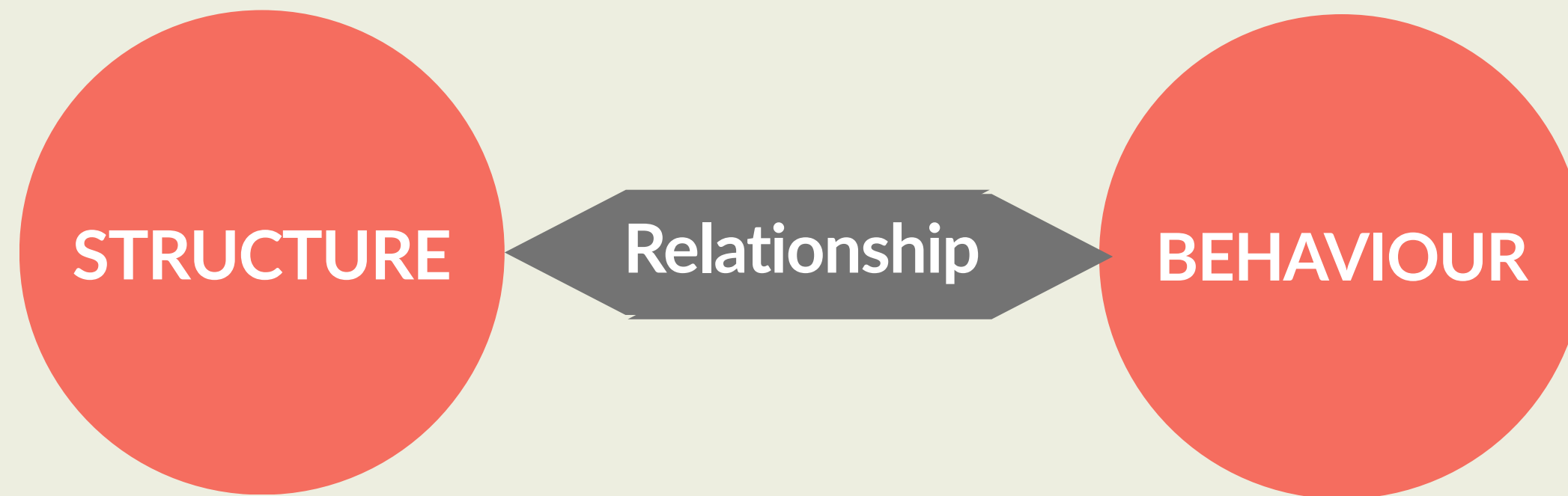
"The hand made the Slinky bounce."
(Attributing behavior entirely to an external trigger).

Everyone focuses on your hand (the external event),
but the true explanation lies in how the Slinky is
constructed internally.

The Systems View

"The hand provided the trigger, but the Slinky's coil design dictated how it behaved."
(Recognizing that internal structure governs response).

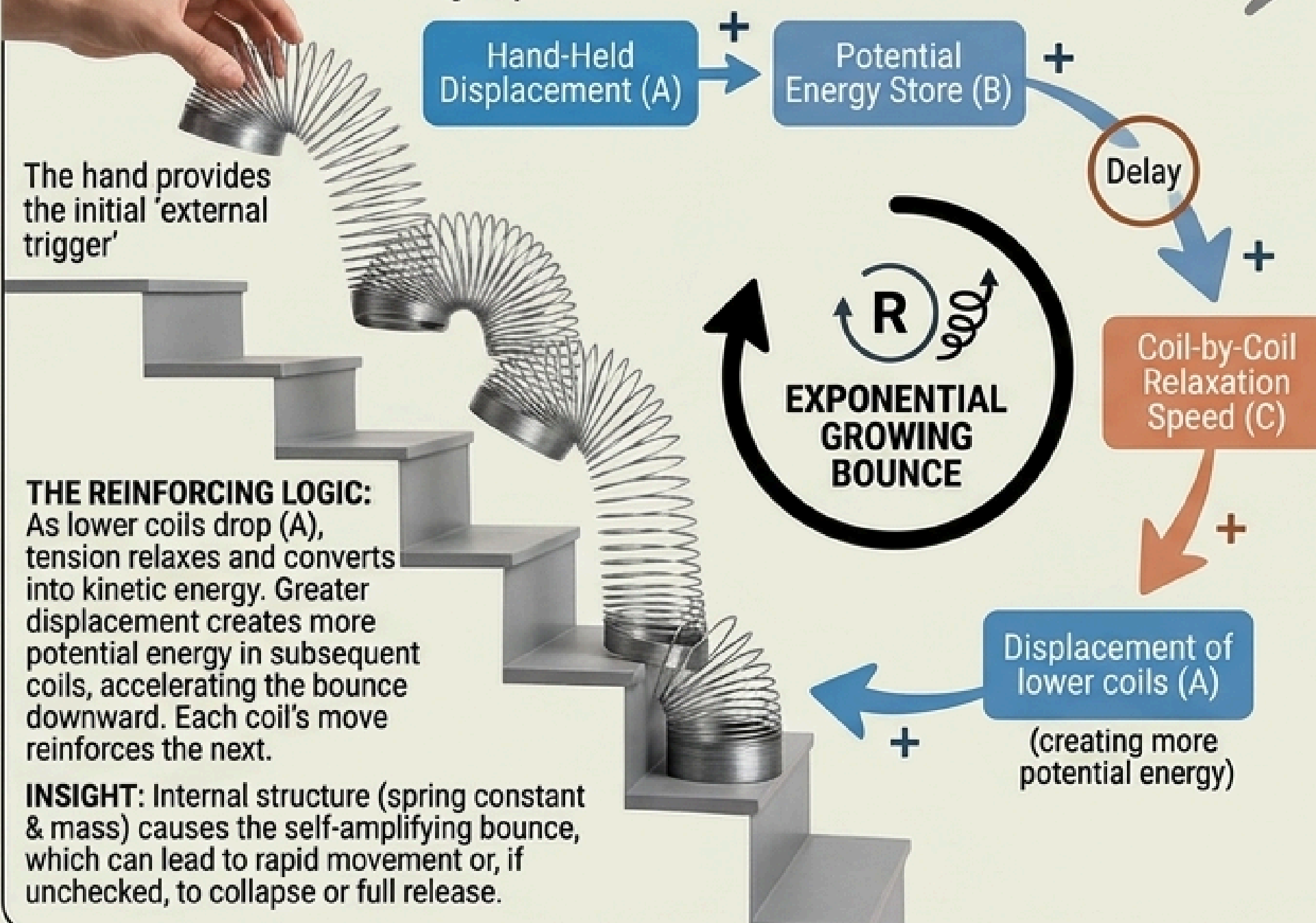
Central insight of the System Theory What Makes a Slinky Bounce?



System behavior is driven by internal structure, not just external force.

PANEL A: REINFORCING LOOP (R) - The Slinky's Progressive Gravity Bounce

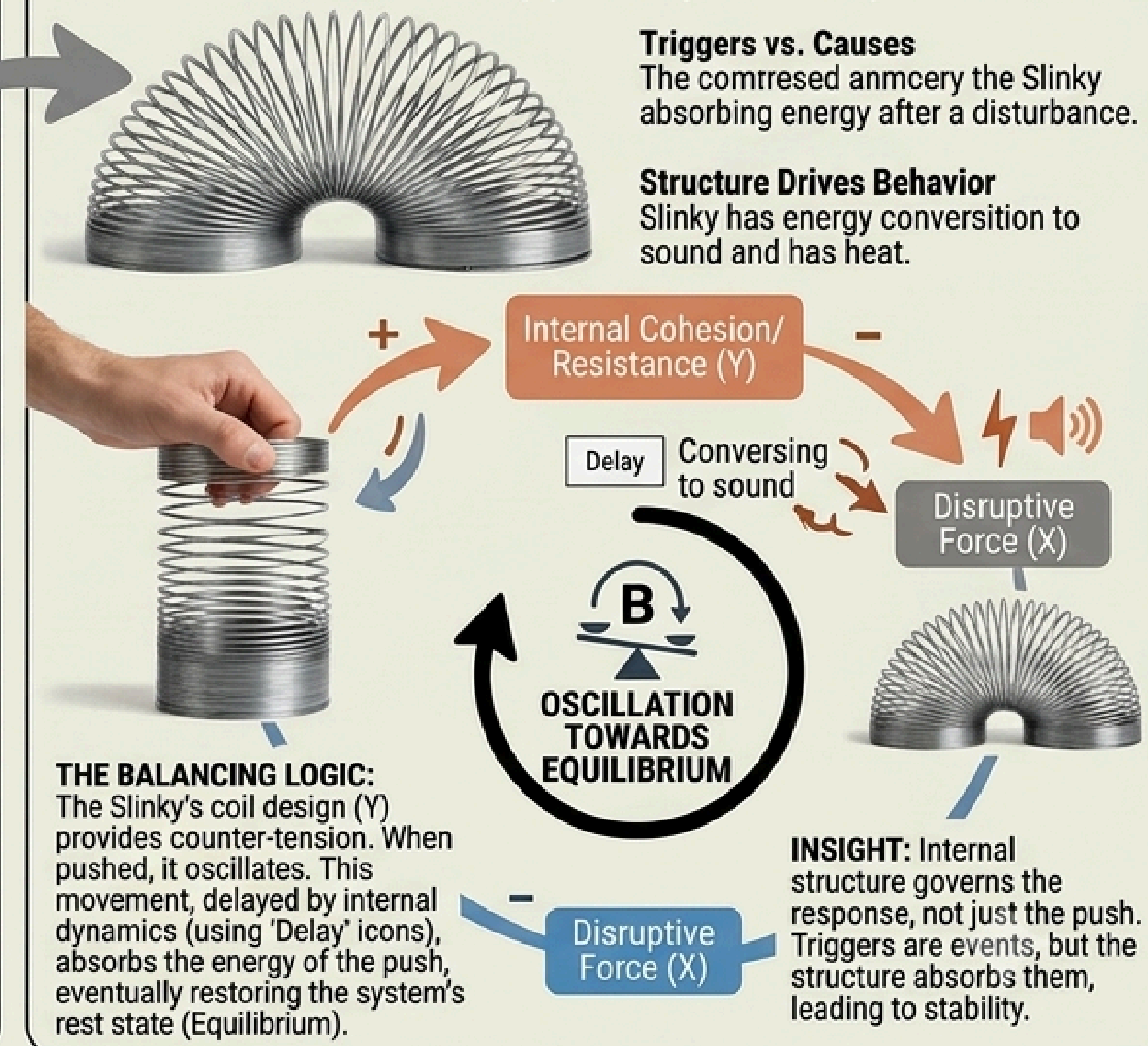
Hand provides the initial 'external trigger' (holding and releasing Slinky) create a clockwise Reinforcing Loop.

**PANEL B: BALANCING LOOP (B) - Absorption and Equilibrium Restoral****Triggers vs. Causes**

The compressed Slinky absorbing energy after a disturbance.

Structure Drives Behavior

Slinky has energy conversion to sound and has heat.



Core System Principles Revealed by the Slinky

Systemic Principle: Structure Drives Behavior.

The Slinky Example: The coils, material, and spring constant determine how the Slinky oscillates when pushed.

Real-World Application: A high-turnover company isn't caused by "bad employees"; the workplace incentive structure naturally breeds burnout.

Core System Principles Revealed by the Slinky

Systemic Principle: Triggers vs. Causes

The Slinky Example: The push (external force) is merely an event trigger. The real cause of the motion is the spring mechanism inside.

Real-World Application: Blaming a market crash solely on a single headline misses the fragile financial leverage built up over years beneath the surface. surface.

Core System Principles Revealed by the Slinky

Systemic Principle: Delays & Feedback

The Slinky Example: When you lift a Slinky, the bottom doesn't move instantly—the tension transfers coil by coil with a visible time delay.

Real-World Application: Policy changes made today in education or green energy take years to show results because feedback travels through structural delays.

Core System Principles Revealed by the Slinky

Systemic Principle: Self-Organisation & Stocks

The Slinky Example: When left alone after a disturbance, the Slinky absorbs energy and eventually returns to its rest state (equilibrium).

Real-World Application: Resilient supply chains use safety stock to absorb demand spikes without collapsing or overreacting.

Key Takeaway

When a system acts poorly, our natural instinct is to blame external forces or individuals (the "push").

Systems thinkers look past the push to ask: "What about the structure of this system makes this behavior predictable?"

LECTURE 3

Assessment Scheme

Individual Assignments - 20 Marks

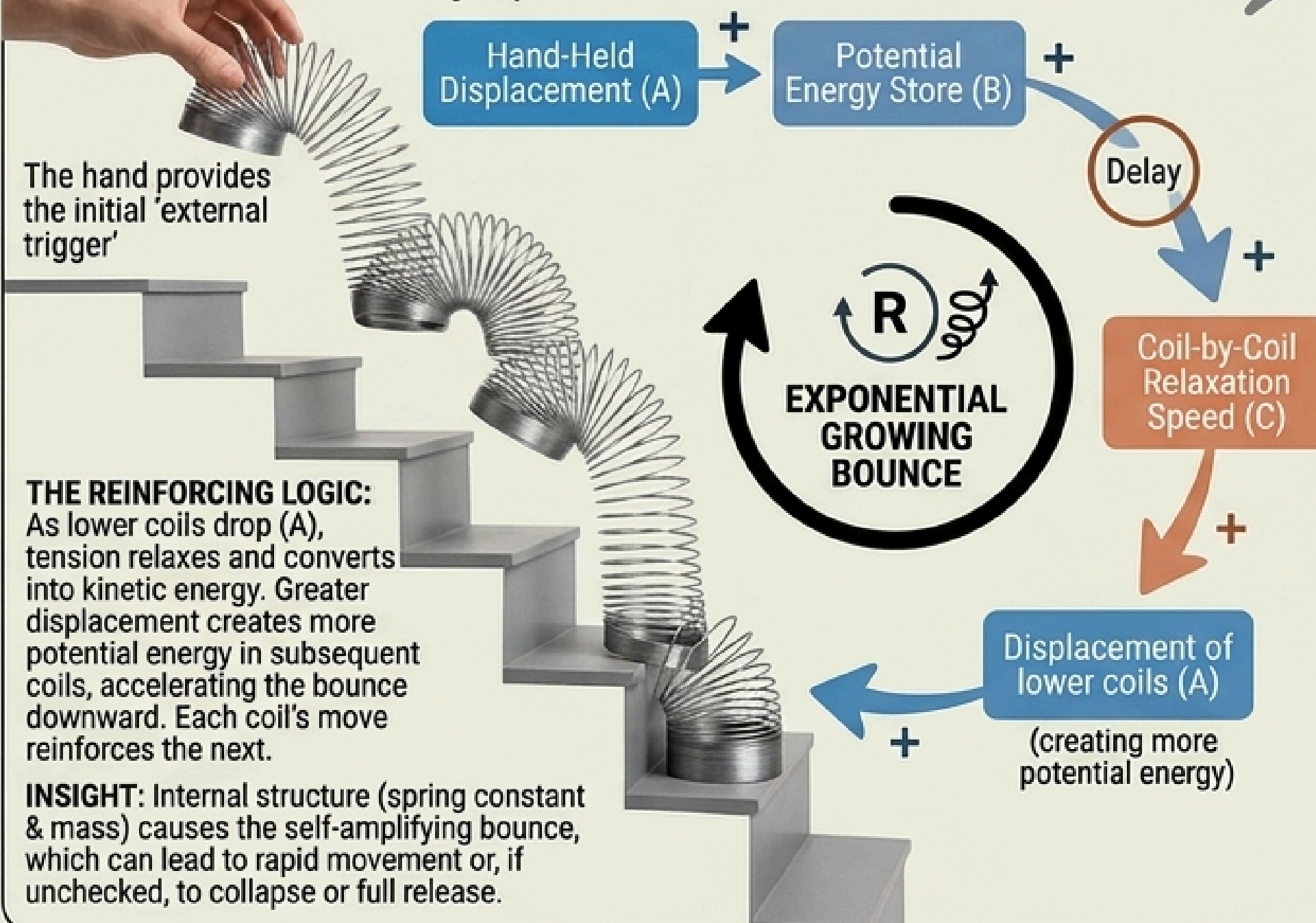
Quiz - 10 Marks

Project - 60 +10 Marks

Assessment - Relative

PANEL A: REINFORCING LOOP (R) - The Slinky's Progressive Gravity Bounce

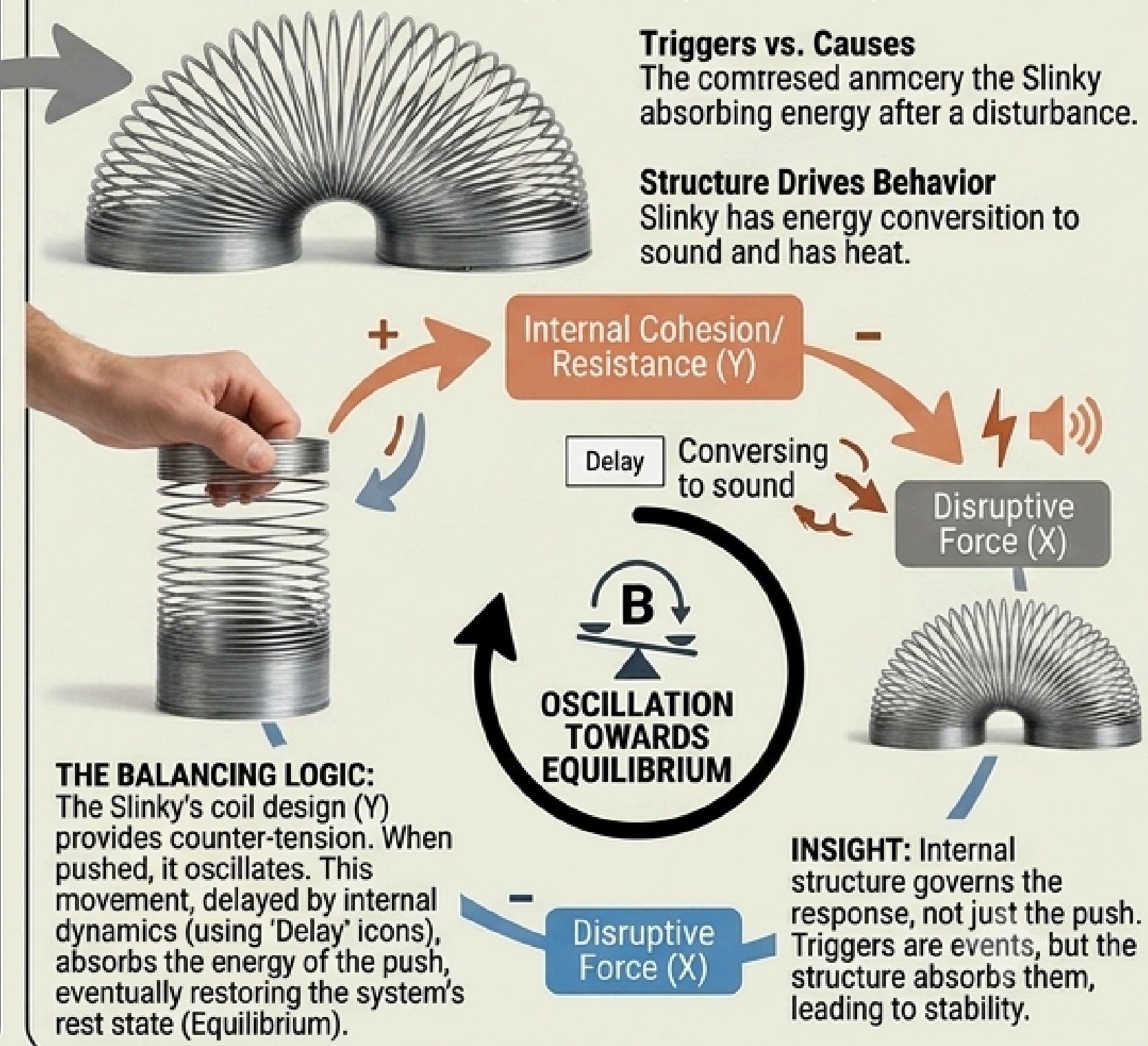
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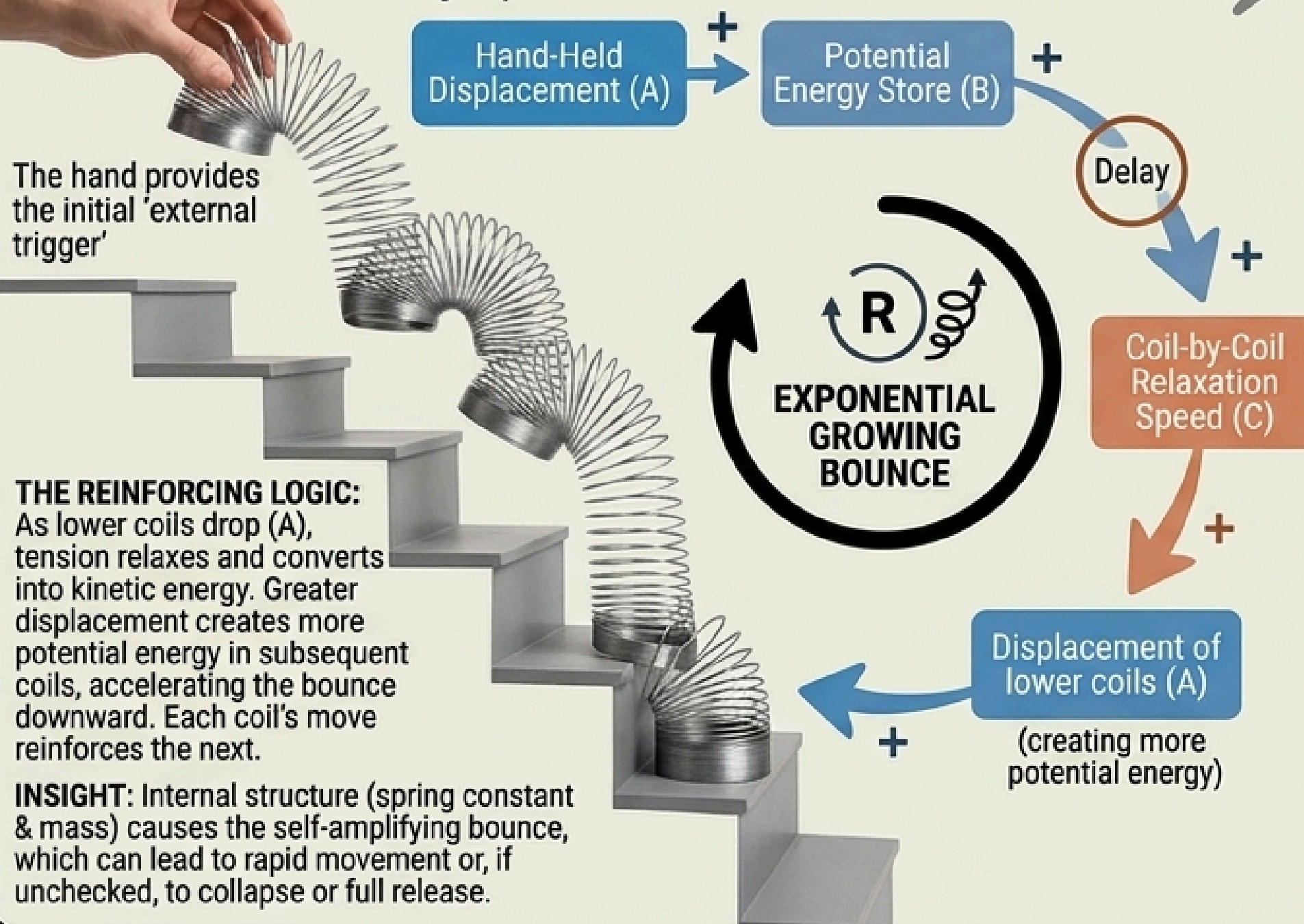
Structure Drives Behavior

Slinky has energy conversion to sound and has heat.



PANEL A: REINFORCING LOOP (R) - The Slinky's Progressive Gravity Bounce

Hand provides the initial 'external trigger' (holding and releasing Slinky) create a clockwise Reinforcing loop.

**PANEL B: BALANCING LOOP (B) - Absorption and Equilibrium Restoral**

This illustrates a Reinforcing Loop (\$R\$). When you release a Slinky down a flight of stairs, your hand provides an initial external trigger. But what keeps it bouncing? The physical structure of the coils converts potential energy to kinetic energy repeatedly. Each movement amplifies the next, causing an exponentially growing step bounce.

In business, reinforcing loops drive rapid growth—or systemic collapse if unchecked.

PANEL A: REINFORCING LOOP (R) - The Slinky's Progressive Gravity Bounce

Hand provides the initial 'external trigger' (holding and releasing Slinky) create a clockwise Reinforcing Loop.

This illustrates a **Balancing Loop (B)**. If you compress a Slinky between your hands and let go, the internal tension creates counter-resistance. The energy dissipates into sound and heat over time, absorbing the shock and restoring equilibrium. Balancing loops provide stability and self-regulation."

As lower coils drop (A), tension relaxes and converts into potential energy, creating more potential energy in subsequent coils, accelerating the bounce downward. Each coil's move reinforces the next.

INSIGHT: Internal structure (spring constant & mass) causes the self-amplifying bounce, which can lead to rapid movement or, if unchecked, to collapse or full release.

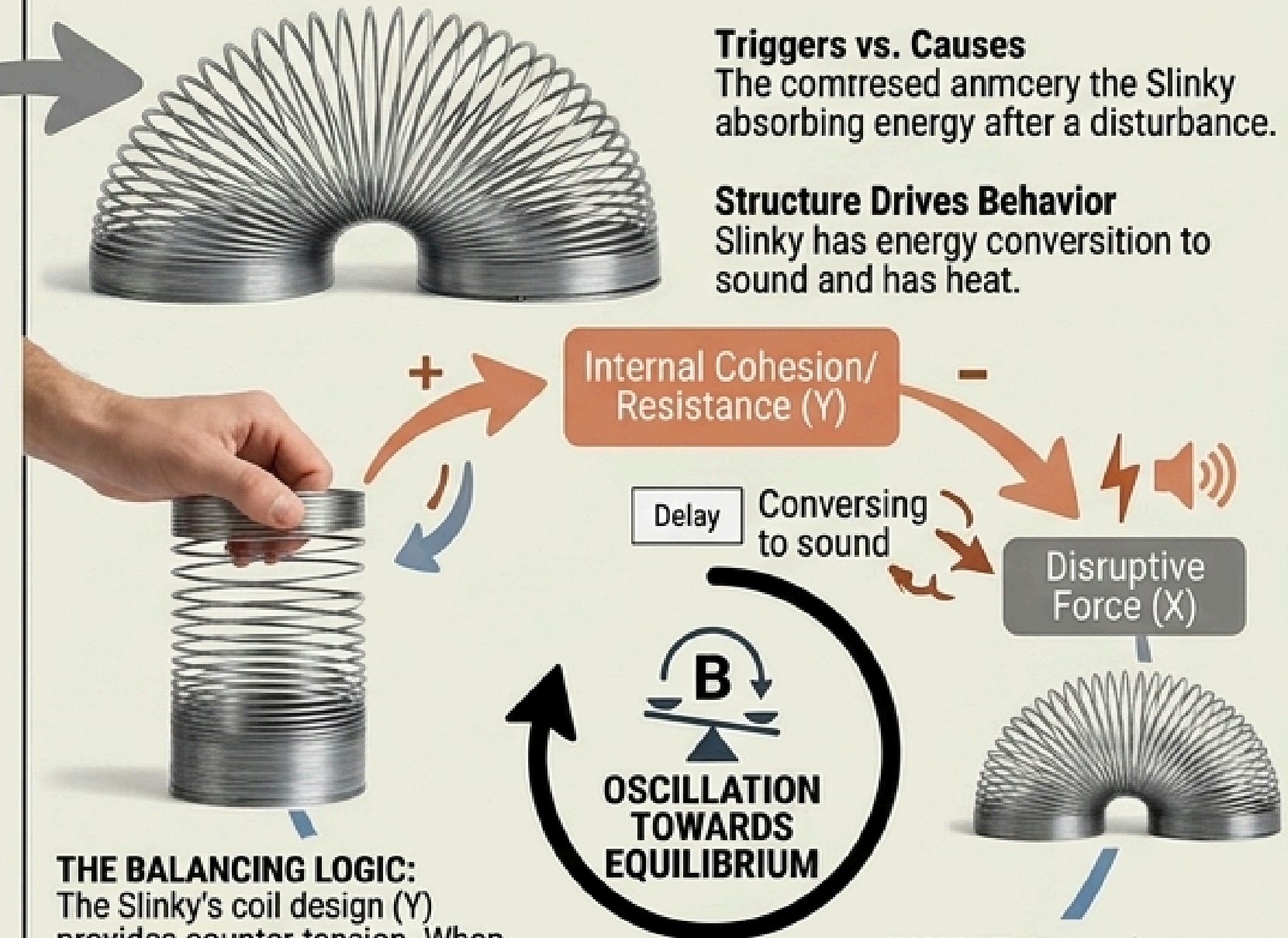
Displacement of lower coils (A)
(creating more potential energy)

PANEL B: BALANCING LOOP (B) - Absorption and Equilibrium Restoral**Triggers vs. Causes**

The compressed Slinky absorbing energy after a disturbance.

Structure Drives Behavior

Slinky has energy conversion to sound and has heat.

**THE BALANCING LOGIC:**

The Slinky's coil design (Y) provides counter-tension. When pushed, it oscillates. This movement, delayed by internal dynamics (using 'Delay' icons), absorbs the energy of the push, eventually restoring the system's rest state (Equilibrium).

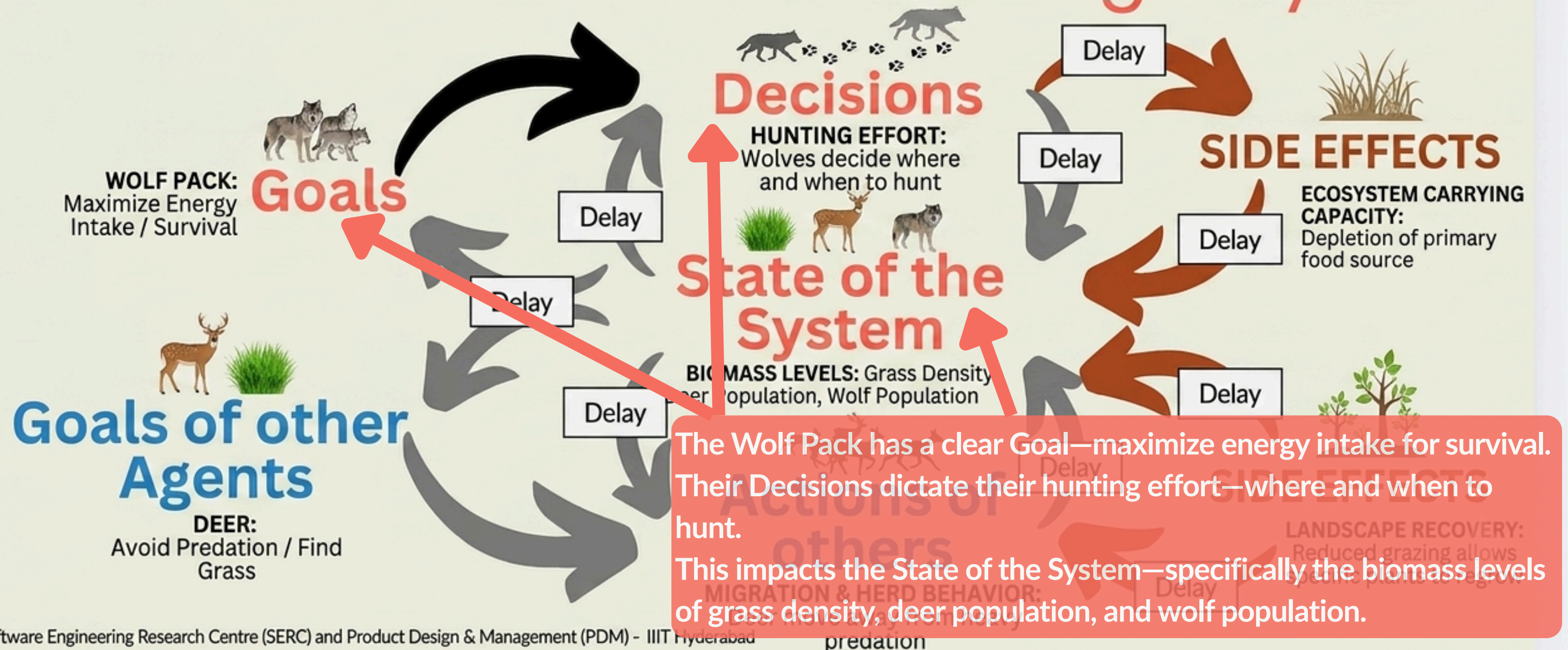
Disruptive Force (X)

INSIGHT: Internal structure governs the response, not just the push. Triggers are events, but the structure absorbs them, leading to stability.

Internal structure governs response, not just the external push. When a system misbehaves, don't just blame the hand that pushed it—look at how the coils are designed.

**Systems Thinking views Food Chain as
a dynamic, self-regulating feedback network
where every organism is an interconnected
node.**

We are embedded in a larger system



Deer are other agents with their own goals—avoid predation and find grass. Their Actions involve moving herds away from heavy predation areas. Notice the Side Effects on the right side of the slide. If wolf predation drops, deer overgraze the land, pushing past the Ecosystem Carrying Capacity. Conversely, when wolves manage the deer population, it creates a delayed side effect: Landscape Recovery, allowing trees and vegetation to regrow.

Goals of other Agents

DEER:
Avoid Predation / Find Grass

State of the System

BIOMASS LEVELS: Grass Density, Deer Population, Wolf Population

Actions of others

MIGRATION & HERD BEHAVIOR:
Deer move away from heavy predation

Delay

Delay

Delay

Delay

Delay

Delay

SIDE EFFECTS

ECOSYSTEM CARRYING CAPACITY:
Depletion of primary food source

SIDE EFFECTS

LANDSCAPE RECOVERY:
Reduced grazing allows specific plants to regrow

We are embedded in a larger system

Notice how every single interaction has an embedded Time Delay. Vegetation doesn't regrow overnight, and wolf populations don't adjust immediately.

This is how natural systems balance themselves—and it's the exact same feedback architecture we must design for in products and organizations.



What Do We Learn About Systems Thinking from the Food Chain?

*The outcomes in a system are governed by
dynamic feedback loops, multi-agent
interactions, and time delays—not by isolated
cause-and-effect actions.*

Interconnectedness & Feedback Loops

- *You cannot change just one element in a system without triggering a chain reaction across the entire network.*
- *The wolf pack's decisions directly affect deer behavior, which in turn impacts grass biomass and ecosystem carrying capacity. Cause and effect loop back on themselves rather than moving in a straight line.*

Multi-Agent Dynamics

- *Every actor within a system (e.g., wolves, deer) is an independent agent driven by its own goals and incentives.*
- *Systems thinkers do not view lower-level actors as passive; they anticipate that other agents will adapt their behavior (e.g., deer moving herds to avoid heavy predation areas), which shifts the overall state of the system.*

Unintended Side Effects & Counterintuitive Behavior

- *Focusing purely on a single goal (e.g., wolves hunting for survival) creates unintended ripple effects elsewhere in the network.*
- *When wolf predation drops, it doesn't just mean "more deer"—it leads to overgrazing that pushes past the ecosystem's carrying capacity, ultimately harming the habitat that sustains all species.*

Time Delays Mask Systemic Causes

- *Responses within complex systems are rarely immediate. Vegetation takes time to regrow, and animal populations take time to adjust.*
- *Because of these embedded time delays, the ultimate consequences of an action (or lack of action) may not be visible until much later, making long-term feedback awareness critical for managing organizations, products, and environments alike.*

Whether managing a natural ecosystem or designing human organizations and products, success requires mapping the entire feedback architecture—accounting for the goals of other agents, delayed consequences, and the underlying carrying capacity of the system.

Three Components of a System

1. **Elements** — the people/nodes (the students themselves)
2. **Interconnections** — the yarn strands (relationships, dependencies, flows)
3. **Purpose/Function** — what emerges from the whole (a class bonded by a shared, tangible experience — even though no one designed that outcome)

Characterstics of a System

1. All systems have a purpose
2. All parts of a system have a function
3. The performance of a system is affected by how the parts are arranged
4. Systems attempt to maintain stability

Elements of a System

- Components – parts/modules
- Relationships – interactions
- Boundary – what's inside/outside
- Environment – external influences
- Feedback Loops – reinforcing/balancing
- Delays – time gaps

*Please note that the purpose of a system isn't
always the stated goal of its parts.*

Think of it like a traffic jam.

*Nobody wakes up in the morning and says, "My goal today is to sit
on the highway for two hours and create a traffic jam!"*

Everyone's individual goal is just to get to work on time.

*Yet, when hundreds of people drive on the same road at the same
time, the structure of the road system creates a massive traffic jam
anyway.*

What People Want <not equal to> What the System Does

- *People have personal intentions (like getting home fast, or just attending a workshop). But the overall system doesn't care about individual intentions—it responds to how its pieces are linked together.*

The "Rules" Drive the Result

- *If you put people into a specific setup or environment (the structure), that setup will force a specific outcome. The outcome "emerges" automatically, almost like magic, just because of how the room or game was set up.*

Stop Blaming the People, Look at the Setup

- *When something goes wrong, our first reaction is to blame someone's bad intentions. Systems thinking teaches us that bad outcomes usually come from flawed setups, not bad people.*